

Vol Tape Regime v2

15 vol and rate indices and 38 ETFs, rank-based features, monotone trees refit yearly; every number on this page is out of sample. As of 2026-10-07. Feb to Jul 2020 excluded.

orange = risk-off side, blue = risk-on side

READING

IN

since 2026-07-14 · rule: OUT when the drawdown probability is above its trailing-756 95th percentile, back IN once below the 80th

90

stay-in dial = 100 minus the trailing percentile of the drawdown probability; the rule goes OUT at 5 or below and returns IN above 20

P(5% DRAWDOWN IN 21D)

0.083

trailing pct 10 · decile 4: historically 26% drew down 5%, mean next month -0.64%

P(5% RALLY IN 21D)

0.024

trailing pct 14 · decile 2: historically 2% rallied 5%, 10% drew down 5%

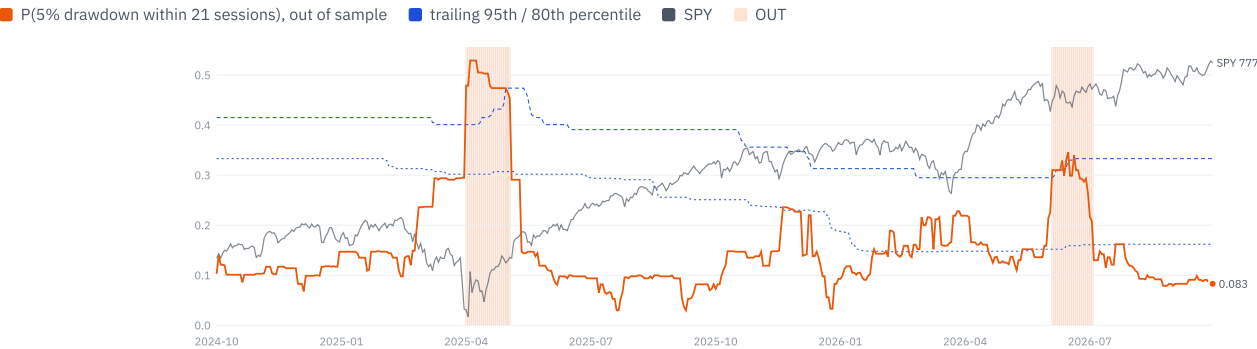
P(VOL EXPANDS 1.5X IN 21D)

0.041

trailing pct 42 · decile 4: historically 9% expanded, mean next month +0.61%

Not inside a vol episode. The drawdown probability sits at the 10th percentile of its own last three years; the rally probability at the 14th; the vol-expansion probability at the 42th. High readings of all three travel together (high vol is a two-tailed state), which is why the stay-in rule keys on the drawdown probability alone and re-enters quickly.

DRAWDOWN PROBABILITY, LAST TWO YEARS, WITH THE TRAILING THRESHOLDS THE RULE USES



WHY TODAY: THE FEATURES INSIDE EACH MODEL AND WHERE THEY SIT AGAINST THEIR OWN HISTORY

drawdown probability · 0.083

XLK:pk5		0.05	+
XLY:pk10		0.08	+
XLK:pk10		0.08	+
XLY:cc10		0.09	+
QQQ:pk10		0.09	+
QQQ:pk5		0.11	+
QQQ:pk21		0.11	+
QQQ:gk10		0.12	+
QQQ:gk5		0.13	+
XLY:pk5		0.16	+

rally probability · 0.024

SPY:pk21		0.11	+
VIX:lv1_rank504		0.14	+
XLV:gk21		0.86	+
SPY:cc10		0.15	+
SPY:cc5		0.17	+
SPY:gk21		0.18	+
SPY:pk10		0.19	+
SPY:cc63		0.20	+
VIX:lv1_rank1260		0.23	+
SPY:pk5		0.25	+

vol-expansion probability · 0.041

IWM:VIX/cc21		0.91	+
IWM:VIX/cc21_rank		0.87	+
SPY:VIX/pk21		0.80	+
SPY:pk21_cc21		0.24	+
SPY:pk21_cc21_rank		0.24	+
SPY:cc21_roc21		0.72	-
SPY:VIX/pk21_rank		0.70	+
SPY:cc21_roc21_rank		0.70	-
SPY:VIX_prem_chg21_rank		0.31	+
QQQ:VXN/cc21		0.68	+

Each probability comes from 15 features chosen inside the last training window. Bars show each feature's trailing-rank distance from its median, signed by the direction the model is constrained to use: orange pushes the probability up, blue pushes it down.

WHAT EACH PROBABILITY MEANT, OUT OF SAMPLE, BY DECILE (2005 TO DATE)

P_OFF DECILE	RANGE	FWD 21D	MEAN DD	P(5% DD)	P(5% UP)	P(VOL X1.5)
1	0.000-0.022	+0.49%	-1.9%	0.098	0.017	0.290
2	0.022-0.038	+0.77%	-1.9%	0.098	0.081	0.158
3	0.038-0.059	+0.89%	-1.7%	0.060	0.055	0.162
4	0.059-0.085	-0.64%	-3.5%	0.258	0.087	0.284
5	0.085-0.102	+1.40%	-1.6%	0.089	0.098	0.183
6	0.102-0.128	+1.31%	-2.2%	0.104	0.185	0.153
7	0.128-0.156	+1.02%	-2.3%	0.090	0.147	0.119
8	0.156-0.241	+0.98%	-2.7%	0.217	0.208	0.068
9	0.241-0.347	+0.82%	-3.6%	0.267	0.279	0.090
10	0.347-0.568	+1.75%	-4.5%	0.347	0.520	0.004

P_ON DECILE	RANGE	FWD 21D	MEAN DD	P(5% DD)	P(5% UP)	P(VOL X1.5)
1	0.000-0.009	+0.87%	-1.4%	0.058	0.036	0.234
2	0.009-0.025	+0.80%	-1.9%	0.104	0.021	0.258
3	0.025-0.041	+0.67%	-1.8%	0.072	0.036	0.209
4	0.041-0.066	+0.79%	-1.9%	0.075	0.041	0.164
5	0.066-0.100	+0.84%	-2.3%	0.143	0.100	0.168
6	0.100-0.132	+1.23%	-2.2%	0.115	0.183	0.121
7	0.132-0.164	+1.07%	-2.4%	0.147	0.177	0.077
8	0.164-0.241	+1.08%	-3.1%	0.258	0.258	0.126
9	0.241-0.426	+0.64%	-3.7%	0.305	0.311	0.073
10	0.428-0.766	+0.79%	-5.4%	0.350	0.512	0.081

P_VOL DECILE	RANGE	FWD 21D	MEAN DD	P(5% DD)	P(5% UP)	P(VOL X1.5)
1	0.000-0.003	+0.65%	-3.5%	0.205	0.215	0.023
2	0.003-0.011	+0.96%	-2.9%	0.207	0.275	0.058
3	0.011-0.025	+0.70%	-2.7%	0.170	0.147	0.053
4	0.025-0.053	+0.61%	-3.0%	0.198	0.177	0.094
5	0.053-0.090	+0.31%	-2.8%	0.194	0.183	0.098
6	0.090-0.138	+0.66%	-2.6%	0.166	0.170	0.115
7	0.138-0.197	+1.30%	-2.3%	0.151	0.130	0.169
8	0.197-0.269	+0.90%	-2.4%	0.155	0.123	0.200
9	0.269-0.407	+1.60%	-1.9%	0.100	0.156	0.220
10	0.407-0.842	+1.08%	-1.9%	0.083	0.100	0.480

TWO-STATE STAY-IN RULES ON THE PROBABILITIES, FULLY OUT OF SAMPLE, 2006 TO DATE

RULE	ANN %	VOL %	SHARPE	MAX DD (LOG)	ULCER %	EXPOSURE	SPELLS	ANN 05-12	ANN 13-19	ANN 20-26	DD 05-12	DD 13-19	DD 20-26	TODAY
SPY buy and hold	10.8	18.13	0.6	-80.3	15.24	1	0	4	13.6	15.2	-80.3	-21.5	-28.1	IN
A: OUT P_vol>q0.85, IN P_vol<q0.5	5.22	15.81	0.33	-77.6	22.35	0.69	91	-2.7	10.2	8.5	-77.6	-21.2	-31	IN
B: OUT P_vol>q0.85 & P_on<q.5, IN P_vol<q0.5 or P_on>q.8	8.05	17.02	0.47	-80.3	20.25	0.76	60	-1.3	12	14.1	-80.3	-21.5	-28.1	IN
A: OUT P_vol>q0.85, IN P_vol<q0.7	7.44	16.4	0.45	-76.3	20.94	0.78	116	-2.5	13.7	11.6	-76.3	-17.7	-30.4	IN
B: OUT P_vol>q0.85 & P_on<q.5, IN P_vol<q0.7 or P_on>q.8	8.48	17.28	0.49	-80.3	22.83	0.82	75	-2.4	13.2	15.4	-80.3	-21.5	-28.1	IN
A: OUT P_vol>q0.90, IN P_vol<q0.5	6.14	16.57	0.37	-93.7	26.4	0.76	67	-3.7	11.4	11.3	-93.7	-20.2	-31	IN
B: OUT P_vol>q0.90 & P_on<q.5, IN P_vol<q0.5 or P_on>q.8	7.76	17.18	0.45	-80.3	19.31	0.8	49	-1	12.5	12.3	-80.3	-21.5	-28.1	IN
A: OUT P_vol>q0.90, IN P_vol<q0.7	8.19	16.94	0.48	-82.6	21.51	0.83	75	-1.6	13.3	13.4	-82.6	-18.3	-30.4	IN
B: OUT P_vol>q0.90 & P_on<q.5, IN P_vol<q0.7 or P_on>q.8	8.36	17.41	0.48	-80.3	21.19	0.85	55	-1.2	13.3	13.5	-80.3	-21.5	-28.1	IN
A: OUT P_vol>q0.95, IN P_vol<q0.5	7.09	17.08	0.42	-80.5	21.65	0.82	48	-2.1	11.6	12.3	-80.5	-21.5	-32.3	IN
B: OUT P_vol>q0.95 & P_on<q.5, IN P_vol<q0.5 or P_on>q.8	7.36	17.44	0.42	-80.3	21.24	0.84	39	-2.1	12.2	12.5	-80.3	-21.5	-28.1	IN
A: OUT P_vol>q0.95, IN P_vol<q0.7	7.89	17.36	0.45	-79.9	22.88	0.88	49	-2.3	12.9	13.7	-79.9	-21.5	-31.6	IN
B: OUT P_vol>q0.95 & P_on<q.5, IN P_vol<q0.7 or P_on>q.8	7.84	17.63	0.44	-80.3	23	0.89	40	-2.5	13	13.7	-80.3	-21.5	-28.1	IN
C: OUT P_off>q0.90, IN P_off<q0.6	9.08	12.22	0.74	-21.8	6.28	0.75	17	7.5	10.2	9.5	-20.6	-16.2	-21.8	IN
C: OUT P_off>q0.90, IN P_off<q0.8	11.38	13.2	0.86	-23.8	6.21	0.84	19	10.8	11.9	11.4	-20.6	-15.7	-23.8	IN
D: OUT P_off>q0.90 & P_on<q.5, IN P_off<q.6 or P_on>q.8	10.8	18.13	0.6	-80.3	15.24	1	0	4	13.6	15.2	-80.3	-21.5	-28.1	IN
E: OUT P_offf63>q0.90, IN P_offf63<q.6	9.51	12.22	0.78	-20.6	7.06	0.75	20	5.6	10.6	12.7	-20.6	-14.1	-19.8	IN
F: OUT P_off>q0.90 & P_offf63>q.8, IN both below q.6	9.53	11.9	0.8	-21.8	5.64	0.73	11	7.5	10.2	11.1	-20.6	-13	-21.8	IN
C: OUT P_off>q0.95, IN P_off<q0.6	9.35	12.81	0.73	-27.4	8.17	0.79	13	6.6	9.7	12	-20.6	-17.7	-27.4	IN
C: OUT P_off>q0.95, IN P_off<q0.8	11.21	13.62	0.82	-25	7.31	0.87	14	9.3	11	13.6	-20.6	-22.5	-25	IN
D: OUT P_off>q0.95 & P_on<q.5, IN P_off<q.6 or P_on>q.8	10.8	18.13	0.6	-80.3	15.24	1	0	4	13.6	15.2	-80.3	-21.5	-28.1	IN
E: OUT P_offf63>q0.95, IN P_offf63<q.6	9.11	13.21	0.69	-28.1	7.22	0.79	15	5.7	10.6	11.3	-20.6	-14.1	-28.1	IN
F: OUT P_off>q0.95 & P_offf63>q.8, IN both below q.6	10.35	12.15	0.85	-21.8	5.78	0.75	8	7.5	10.2	13.7	-20.6	-13	-21.8	IN
C: OUT P_off>q0.98, IN P_off<q0.6	9.6	13.74	0.7	-28.1	8.54	0.83	9	5.8	10	13.5	-22.9	-15.8	-28.1	IN
C: OUT P_off>q0.98, IN P_off<q0.8	11.37	14.35	0.79	-28.1	7.75	0.89	9	8.5	11.2	14.8	-20.9	-22.5	-28.1	IN
D: OUT P_off>q0.98 & P_on<q.5, IN P_off<q.6 or P_on>q.8	10.8	18.13	0.6	-80.3	15.24	1	0	4	13.6	15.2	-80.3	-21.5	-28.1	IN
E: OUT P_offf63>q0.98, IN P_offf63<q.6	10.3	13.61	0.76	-28.1	6.73	0.84	12	5.7	12.6	12.9	-20.6	-12.7	-28.1	IN
F: OUT P_off>q0.98 & P_offf63>q.8, IN both below q.6	10.31	13.39	0.77	-28.1	6.34	0.8	7	7.5	10.3	13.5	-20.6	-13	-28.1	IN

Blue row: the reference rule shown above. Orange row: the rule that was pre-registered before the grid was run (vol expansion likely with no rebound setup) and failed, because an expected vol expansion is a low-drawdown state out of sample. The reference rule was chosen from the pre-specified grid after the grid was run, and its re-entry threshold was moved from the 60th to the 80th percentile after the slow version sat out 2009 and 2016; both versions are in the table. Drawdowns are in log units (-80 log = -55% price).

YEAR	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	20
book %	13.4	11	-8.8	19	14	1.9	14.8	28	12.6	-7.3	7.6	19.6	-5.9	21.9	14.3	25.3	-17	23.3	22.2	5.1	1
SPY %	14.7	5	-45.9	23.4	14	1.9	14.8	28	12.6	1.2	11.3	19.6	-4.7	27.2	14.3	25.3	-20.1	23.3	22.2	16.3	1
diff	-1.3	+6.0	+37.1	-4.4	0.0	0.0	0.0	0.0	0.0	-8.5	-3.7	0.0	-1.2	-5.3	0.0	0.0	+3.1	0.0	0.0	-11.2	-2

OUT spells of the reference rule (SPY move while out)

2006-07-28 to 2006-08-15 · 13d · +0.5%

2007-10-17 to 2007-10-18 · 2d · -0.4%

2007-10-25 to 2007-10-31 · 5d · +1.8%

2007-11-09 to 2007-11-20 · 8d · -0.3%

2008-01-23 to 2009-06-02 · 343d · -31.4%

2015-08-26 to 2015-11-30 · 67d · +7.6%

2016-03-01 to 2016-03-21 · 15d · +3.8%
2018-02-09 to 2018-03-12 · 21d · +6.3%
2018-03-27 to 2018-05-01 · 25d · +1.7%
2018-10-24 to 2019-01-28 · 64d · 0.0%
2022-02-24 to 2022-04-04 · 28d · +6.8%
2022-05-04 to 2022-07-18 · 51d · -11.2%
2025-04-07 to 2025-05-08 · 23d · +11.4%
2026-06-11 to 2026-07-13 · 21d · +1.8%

ADD AS A SLEEVE (LONG ONLY ON ADD SESSIONS, RULE IN; EX-2020)	ANN %	SHARPE	MAX DD	SESSIONS LONG	SHARE OF IN SESSIONS	MEAN DAILY BP WHILE LONG	SPELLS
sleeve ADD: P_on >= q.80	2.84	0.4	-15.7	610	0.14	9.4	89
sleeve ADD: P_on >= q.90	1.68	0.29	-12.5	342	0.08	9.9	52
sleeve ADD: P_on >= q.80, held until < q.60	3.96	0.45	-19.4	969	0.22	8.3	55
sleeve ADD: P_on >= q.90, held until < q.60	2.49	0.31	-18.2	692	0.16	7.3	31
sleeve: all IN sessions (unconditional)	11.21	0.82	-25	4411	1	5.1	16

ADD AS A NO-LEVERAGE BOOK (BASE WEIGHT WHEN IN, 100% ON ADD, 0 WHEN OUT; EX-2020)	ANN %	SHARPE	MAX DD	MEAN WEIGHT
reference: 100% when IN, 0 when OUT	11.21	0.82	-25	0.87
SPY buy and hold	10.8	0.6	-80.3	1
75% when IN, no ADD (the reserve never deployed)	8.4	0.82	-18.8	0.65
75% when IN, 100% on ADD: P_on >= q.80	9.11	0.81	-20.8	0.68
75% when IN, 100% on ADD: P_on >= q.90	8.82	0.81	-20	0.67
75% when IN, 100% on ADD: P_on >= q.80, held until < q.60	9.39	0.8	-21.7	0.7
75% when IN, 100% on ADD: P_on >= q.90, held until < q.60	9.03	0.79	-21.2	0.68
50% when IN, no ADD (the reserve never deployed)	5.6	0.82	-12.5	0.43
50% when IN, 100% on ADD: P_on >= q.80	7.02	0.76	-19.1	0.49
50% when IN, 100% on ADD: P_on >= q.90	6.44	0.76	-17.5	0.47
50% when IN, 100% on ADD: P_on >= q.80, held until < q.60	7.58	0.74	-21	0.53
50% when IN, 100% on ADD: P_on >= q.90, held until < q.60	6.85	0.71	-19.9	0.5

ADD as its own book. The sessions the ADD condition selects (rally probability at or above its trailing 80th percentile, held until it falls below the 60th) earn 8.3 basis points a day against 5.1 for all IN sessions, on 22% of them. There is no free lunch without leverage: a book that holds 75% when IN and goes to 100% only on ADD makes 9.39% against 8.4% for the reserve never deployed and 11.21% for always full. ADD says where a marginal dollar is best put to work; it does not beat being fully invested.

RULE	EX-2020 ANN %	MAX DD LOG	PRICE %	INCL-2020 ANN %	MAX DD LOG	PRICE %	SHARE OF FEB 19 TO MAR 23 2020 TAKEN	FIRST OUT IN THE CRASH	BACK IN	2020 BOOK %	2020 SPY %
SPY buy and hold	10.8	-80.3	-55.2	10.66	-80.3	-55.2	1	never	never OUT	16.8	16.8
C: OUT P_off>q0.90, IN P_off<q0.6	9.08	-21.8	-19.6	8.67	-21.8	-19.6	0.22	2020-03-02 (SPY -8.6%)	2020-08-13	-3.1	16.8
C: OUT P_off>q0.90, IN P_off<q0.8	11.38	-23.8	-21.2	10.97	-23.8	-21.2	0.22	2020-03-02 (SPY -8.6%)	2020-06-02	10.1	16.8
C: OUT P_off>q0.95, IN P_off<q0.6	9.35	-27.4	-24	8.93	-27.4	-24	0.22	2020-03-02 (SPY -8.6%)	2020-08-13	7.2	16.8
C: OUT P_off>q0.95, IN P_off<q0.8	11.21	-25	-22.2	11.04	-25	-22.2	0.22	2020-03-02 (SPY -8.6%)	2020-06-02	16.6	16.8
F: OUT P_off>q0.95 & P_off<q.8, IN both below q.6	10.35	-21.8	-19.6	9.91	-21.8	-19.6	0.22	2020-03-02 (SPY -8.6%)	2020-08-14	7.2	16.8
C: OUT P_off>q0.98, IN P_off<q0.6	9.6	-28.1	-24.5	9.03	-28.1	-24.5	0.29	2020-03-03 (SPY -11.3%)	2020-08-13	4.3	16.8
C: OUT P_off>q0.98, IN P_off<q0.8	11.37	-28.1	-24.5	11.06	-28.1	-24.5	0.29	2020-03-03 (SPY -11.3%)	2020-06-02	13.7	16.8

The 2020 path. Every statistic on this page excludes Feb to Jul 2020 by design; here the frozen rules, fits and thresholds unchanged, are scored with the window included. Reference rule: max drawdown -25 log excluding the window and -25 log including it; it went OUT 2020-03-02 (SPY -8.6%), took 0.22 of the Feb 19 to Mar 23 decline, and came back IN 2020-06-02.

15% DECLINES: THE LOSS THAT MATTERS

P(15% DECLINE WITHIN 63 SESSIONS)
0.009
trailing pct 67 · decile 7: historically 3% saw one (base 5%), mean worst close within 63d -4.4%

P(15% DECLINE WITHIN 126 SESSIONS)
0.100
trailing pct 64 · decile 8: historically 6% saw one (base 11%)

DEDICATED 15%/63 MODEL, OUT OF SAMPLE
0.723
AUC by era 0.750 / 0.820 / 0.550 · year-bootstrap CI 0.587-0.830 · top quintile hit 0.129

PERMUTATION TEST, 15%/63 PIPELINE
0.724 vs 0.530
real AUC vs the mean of 20 null pipelines (sd 0.115, max 0.780) · z 1.7

Of the 11 declines: drawdown probability (5%/21): 3 already extreme at the peak, 1 crossed before SPY was 5% down, 4 crossed later, 3 never. Dedicated 15%/63 probability: 3 already extreme at the peak, 1 crossed before SPY was 5% down, 4 crossed later, 3 never. Vol-expansion probability: 0 already extreme at the peak, 3 crossed before SPY was 5% down, 2 crossed later, 6 never. The "already extreme" cases are the 2008-09 legs, where the previous leg had put every probability at its ceiling.

PEAK	TROUGH	DEPTH %	SESSIONS	P_OFF (5%/21)	PCT AT PEAK	FIRST ABOVE 95TH	P15/63 (NEW)	PCT AT PEAK	FIRST ABOVE 95TH	P_VOL	PCT AT PEAK	FIRST ABOVE 95TH
2007-10-09	2008-03-10	-17.8	104		0.560	2007-10-17 (+6s, SPY -1.4%)		0.820	2008-01-02 (+58s, SPY -6.9%)		0.280	2007-10-17 (+6s, SPY -1.4%)
2008-05-19	2008-10-10	-37.5	101		0.870	2008-09-18 (+85s, SPY -15.6%)		0.920	2008-09-10 (+79s, SPY -13.1%)		0.680	never
2008-10-13	2008-10-27	-17.2	10		1.000	2008-10-13 (+0s, SPY 0.0%)		0.990	2008-10-13 (+0s, SPY 0.0%)		0.100	never
2008-11-04	2008-11-20	-24.9	12		1.000	2008-11-04 (+0s, SPY 0.0%)		0.990	2008-11-04 (+0s, SPY 0.0%)		0.120	never
2009-01-06	2009-03-09	-27.1	42		1.000	2009-01-06 (+0s, SPY 0.0%)		1.000	2009-01-06 (+0s, SPY 0.0%)		0.510	never
2010-04-23	2010-07-02	-15.7	49		0.500	never		0.160	never		0.890	2010-05-06 (+9s, SPY -7.3%)
2011-04-29	2011-10-03	-18.6	108		0.260	never		0.510	2011-08-19 (+78s, SPY -17.0%)		0.780	2011-05-05 (+4s, SPY -2.1%)
2018-09-20	2018-12-24	-19.3	65		0.540	2018-10-24 (+24s, SPY -9.2%)		0.850	2018-09-27 (+5s, SPY -0.5%)		0.720	never
2022-01-03	2022-06-16	-23	114		0.670	2022-02-24 (+36s, SPY -10.3%)		0.050	2022-03-07 (+43s, SPY -12.2%)		0.130	2022-01-25 (+15s, SPY -9.1%)
2022-08-16	2022-10-12	-16.7	40		0.710	never		0.050	never		0.530	never
2025-02-19	2025-04-08	-18.8	34		0.050	2025-04-07 (+33s, SPY -17.5%)		0.260	never		0.830	2025-02-26 (+5s, SPY -3.0%)

Every SPY peak-to-trough decline of 15% or more since 2005 (zigzag on closes; a 10% rebound or a new high closes an event; 2020 excluded). For each out-of-sample probability: where it sat against its own trailing three years at the peak, and the first session at or after the peak where it crossed its trailing 95th percentile, with the decline already in place by then.

15%/63 probability · 0.009 · features inside the last fit and where they sit (orange pushes up)

XPH:rel63_rank

XSD:amihud21_rank

XHB:dd252

KBE:volvol21_rank

KBE:dd63

KCE:dd63

XHB:dd63

KCE:rel63

KCE:roc63

XSD:roc63

0.01

0.01

0.09

0.09

0.11

0.18

0.19

0.24

0.24

+

+

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P15/63 DECILE	RANGE	P(15% DECLINE IN 63D)	MEAN WORST CLOSE 63D	MEAN FWD 63D
1	0.000-0.000	0.027	-4.0%	+1.2%
2	0.000-0.001	0.008	-2.7%	+2.5%
3	0.001-0.001	0.034	-5.0%	+0.3%
4	0.001-0.003	0.006	-2.7%	+3.8%
5	0.003-0.005	0.004	-3.1%	+3.5%
6	0.005-0.006	0.057	-4.1%	+3.7%
7	0.006-0.011	0.031	-4.4%	+3.4%
8	0.011-0.027	0.006	-6.0%	+0.8%
9	0.027-0.082	0.153	-7.6%	+2.1%
10	0.085-0.589	0.105	-5.6%	+4.7%

RULE	ANN %	SHARPE	MAX DD (LOG)	EXPOSURE	SPELLS	FALSE	MEAN TAKEN	WORST	AVOIDED INSIDE 15% EVENTS	GIVEN UP OUTSIDE	TAKEN PER EVENT (11)
E: OUT P_off63>q0.90, IN P_off63<q.6	9.51	0.78	-20.6	0.75	20	13	0.35	1	170.9	197	0.23 0.00 0.00 0.00 0.00 1.00 1.00 0.45 0.74 0.00 0.47
F: OUT P_off>q0.90 & P_off63>q.8, IN both below q.6	9.53	0.8	-21.8	0.73	11	6	0.35	1	174.2	200	0.15 0.00 0.00 0.00 0.00 1.00 1.00 0.45 0.31 0.00 0.92
I: OUT P_off>q0.9 or P15/63>q0.9, IN both<q0.8	8.56	0.73	-28.1	0.75	34	20	0.35	1	176.8	222.2	0.13 0.00 0.00 0.00 0.00 1.00 0.91 0.03 0.30 1.00 0.46
F: OUT P_off>q0.95 & P_off63>q.8, IN both below q.6	10.35	0.85	-21.8	0.75	8	3	0.36	1	171.4	180.5	0.15 0.00 0.00 0.00 0.00 1.00 1.00 0.45 0.42 0.00 0.92
C: OUT P_off>q0.90, IN P_off<q0.6	9.08	0.74	-21.8	0.75	17	9	0.38	1	170.7	205.5	0.06 0.00 0.00 0.00 0.00 1.00 1.00 0.28 0.00 0.90 0.92
G: OUT P15/63>q0.85, IN P15/63<q0.7	9.44	0.77	-30	0.77	43	26	0.4	1.28	159.1	186.7	0.27 0.22 0.00 0.00 0.00 1.00 1.28 0.00 0.20 1.00 0.47
G: OUT P15/63>q0.9, IN P15/63<q0.8	10.53	0.81	-27.2	0.83	29	13	0.42	1.32	156.7	162.1	0.24 0.15 0.00 0.00 0.00 1.00 1.32 0.03 0.39 1.00 0.46
G: OUT P15/63>q0.9, IN P15/63<q0.7	10.06	0.78	-27.2	0.82	25	13	0.42	1.32	156.3	171.3	0.27 0.15 0.00 0.00 0.00 1.00 1.32 0.00 0.41 1.00 0.46
I: OUT P_off>q0.95 or P15/63>q0.95, IN both<q0.8	10.2	0.81	-28.1	0.82	19	9	0.42	1	160.9	173	0.13 0.00 0.00 0.00 0.00 1.00 0.91 0.21 0.42 1.00 0.92

C: OUT P_off>q0.90, IN P_off<q0.8	11.38	0.86	-23.8	0.84	19	8	0.43	1	156.9	145.2	0.06	0.00	0.00	0.00	0.00	1.00	1.00	0.28	0.49	1.00	0.92
J: OUT P_off>q0.9 and P15/63>q0.8, IN P_off<q0.8	11.32	0.85	-23.8	0.86	14	5	0.44	1	154.5	144	0.18	0.00	0.00	0.00	0.00	1.00	1.00	0.28	0.49	1.00	0.92
C: OUT P_off>q0.95, IN P_off<q0.6	9.35	0.73	-27.4	0.79	13	5	0.47	1	148.8	178.3	0.47	0.00	0.00	0.00	0.00	1.00	1.00	0.45	0.42	0.90	0.92
H: OUT P15/126>q0.85, IN P15/126<q0.7	8.53	0.69	-26.9	0.78	29	16	0.47	1.1	145.4	191.4	0.06	0.17	0.00	0.00	0.00	1.00	0.54	1.00	0.33	1.10	0.92
K: OUT P15/63>q0.9 and P15/126>q0.8, IN both<q0.8	10.14	0.77	-26.5	0.84	14	5	0.47	1	144.3	157.8	0.24	0.13	0.00	0.00	0.00	1.00	1.00	1.00	0.39	1.00	0.46
C: OUT P_off>q0.95, IN P_off<q0.8	11.21	0.82	-25	0.87	14	5	0.5	1	142.3	134.1	0.47	0.00	0.00	0.00	0.00	1.00	1.00	0.45	0.61	1.00	0.92
SPY buy and hold	10.8	0.6	-80.3	1	0	0	1	1	0	0	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00

The fourteen rules that took the least of the eleven 15% declines (events: 2007-10-09 to 2008-03-10 -17.8%; 2008-05-19 to 2008-10-10 -37.5%; 2008-10-13 to 2008-10-27 -17.2%; 2008-11-04 to 2008-11-20 -24.9%; 2009-01-06 to 2009-03-09 -27.1%; 2010-04-23 to 2010-07-02 -15.7%; 2011-04-29 to 2011-10-03 -18.6%; 2018-09-20 to 2018-12-24 -19.3%; 2022-01-03 to 2022-06-16 -23.0%; 2022-08-16 to 2022-10-12 -16.7%; 2025-02-19 to 2025-04-08 -18.8%), plus the reference rule and buy-and-hold. "taken" 1.0 = took the whole decline, 0 = sidestepped it, above 1 = whipsawed into more than the decline. Totals are log-return points over 2006 to date. Every rule that sidesteps more of the declines gives up more outside them; none clears buy-and-hold by more than the reference rule does.

PRICE TRIGGER		RE-ENTRY	CONTROL	ANN %	SHARPE	MAX DD	SPELLS	FALSE	TAKEN	WITH VOL ALERT	ANN %	SHARPE	MAX DD	SPELLS	FALSE	TAKEN
3% below 63d high	IN calm (P_off<q.8 & P_vol<q.7)			5.73	0.56	-27.2	391	266	0.31		4.87	0.4	-38	262	182	0.49
		IN new 42d high		3.86	0.47	-19.3	81	59	0.15		4.68	0.42	-34.4	57	40	0.32
3% below 63d high	IN calm or new 42d high			5.67	0.53	-35.9	406	277	0.32		5.35	0.42	-37.8	267	185	0.53
5% below 63d high	IN calm (P_off<q.8 & P_vol<q.7)			7.35	0.64	-28.5	197	111	0.38		7.49	0.59	-33.5	124	67	0.52
		IN new 42d high		5.33	0.53	-26.7	43	27	0.25		5.77	0.47	-30.6	30	18	0.42
5% below 63d high	IN calm or new 42d high			7.55	0.64	-37.7	204	116	0.4		7.81	0.59	-32.5	125	67	0.55
7% below 63d high	IN calm (P_off<q.8 & P_vol<q.7)			8.3	0.69	-25.5	95	37	0.42		8.73	0.67	-31.3	57	21	0.54
		IN new 42d high		6.84	0.61	-33.5	28	14	0.3		8.13	0.64	-33.7	17	8	0.44
7% below 63d high	IN calm or new 42d high			8.82	0.71	-28	99	40	0.44		9.01	0.67	-33.7	58	21	0.58
10% below 63d high	IN calm (P_off<q.8 & P_vol<q.7)			9.36	0.73	-30.7	43	9	0.48		10.07	0.74	-33.3	25	4	0.62
10% below 63d high	IN new 42d high			10.05	0.79	-33.4	13	1	0.42		10.68	0.77	-35.6	7	0	0.58
10% below 63d high	IN calm or new 42d high			9.81	0.74	-31.5	45	10	0.51		10.2	0.72	-35.6	26	4	0.66

Onset rules: a vol-expansion alert (P_vol above its trailing 90th percentile at some point in the last 42 sessions) that becomes OUT only once SPY sits a given percent below its 63-session closing high, against the same price trigger with no vol filter. Re-entry on calmed probabilities, on a new 42-session closing high, or on either. "taken" is the mean share of the eleven 15% declines the book took; "false" counts OUT spells not followed by any 10% decline.

ROBUSTNESS: MODEL CAPACITY, RAW SINGLE FEATURES, YEAR-BLOCK BOOTSTRAP, AND A PERMUTATION TEST OF THE WHOLE PIPELINE

TARGET	MODEL	AUC 05-12	AUC 13-19	AUC 20-26	AUC ALL	HIT RATE BY PREDICTED QUINTILE (LOW TO HIGH)
P_off	K150 d3 free (the overfit reference)	0.491	0.493	0.629	0.533	0.14 · 0.15 · 0.16 · 0.19 · 0.17
P_off	K30 d2 mono	0.65	0.524	0.671	0.631	0.09 · 0.15 · 0.14 · 0.13 · 0.30
P_off	K15 d2 mono (production)	0.624	0.586	0.716	0.636	0.10 · 0.16 · 0.10 · 0.15 · 0.31
P_off	K8 d1 mono (additive)	0.608	0.556	0.695	0.625	0.10 · 0.15 · 0.09 · 0.18 · 0.29
P_off	K5 d1 mono	0.575	0.548	0.687	0.61	0.12 · 0.15 · 0.08 · 0.16 · 0.30
P_vol	K30 d2 mono	0.625	0.826	0.84	0.751	0.04 · 0.07 · 0.10 · 0.17 · 0.37
P_vol	K15 d2 mono (production)	0.65	0.795	0.819	0.748	0.04 · 0.07 · 0.11 · 0.18 · 0.35
P_vol	K8 d1 mono	0.659	0.818	0.828	0.761	0.04 · 0.07 · 0.10 · 0.18 · 0.37
P_on	K30 d2 mono	0.808	0.798	0.767	0.796	0.03 · 0.03 · 0.13 · 0.23 · 0.42
P_on	K15 d2 mono (production)	0.798	0.787	0.751	0.788	0.03 · 0.04 · 0.14 · 0.22 · 0.41
P_on	K8 d1 mono	0.799	0.783	0.739	0.784	0.02 · 0.04 · 0.14 · 0.22 · 0.41
P_off	raw VIX:liv_rank504	0.678	0.668	0.602	0.66	0.09 · 0.09 · 0.15 · 0.16 · 0.33
P_off	raw VIX3M:liv_rank252	0.653	0.65	0.662	0.658	0.10 · 0.10 · 0.09 · 0.21 · 0.31
P_off	raw SPY:cc21_rank	0.684	0.589	0.667	0.66	0.07 · 0.10 · 0.14 · 0.20 · 0.29
P_off	raw HYG:pk21_rank	0.631	0.598	0.653	0.633	0.10 · 0.11 · 0.14 · 0.15 · 0.31
P_off	raw XLV:rel63	0.592	0.651	0.662	0.616	0.12 · 0.12 · 0.15 · 0.14 · 0.29
P_off	raw XS:avgcorr21_rank	0.549	0.572	0.603	0.58	0.12 · 0.15 · 0.14 · 0.18 · 0.23
P_off	raw -HYG:signedvol21	0.584	0.656	0.655	0.616	0.09 · 0.16 · 0.12 · 0.19 · 0.26
P_off	raw VIX:roc21_rank	0.513	0.562	0.55	0.536	0.17 · 0.13 · 0.13 · 0.17 · 0.21
P_off	raw VIX:roc5_rank	0.505	0.551	0.522	0.521	0.16 · 0.17 · 0.13 · 0.15 · 0.28
P_off	raw -SPY:VIX/cc21_rank	0.596	0.537	0.629	0.594	0.10 · 0.12 · 0.18 · 0.17 · 0.24

P_OFF PERMUTATION TEST

0.636 vs 0.497

real pooled AUC vs the mean of 20 null pipelines (sd 0.048, max 0.612) · z 2.9

P_VOL PERMUTATION TEST

0.748 vs 0.500

real pooled AUC vs the mean of 20 null pipelines (sd 0.025, max 0.533) · z 10.0

P_ON PERMUTATION TEST

0.789 vs 0.508

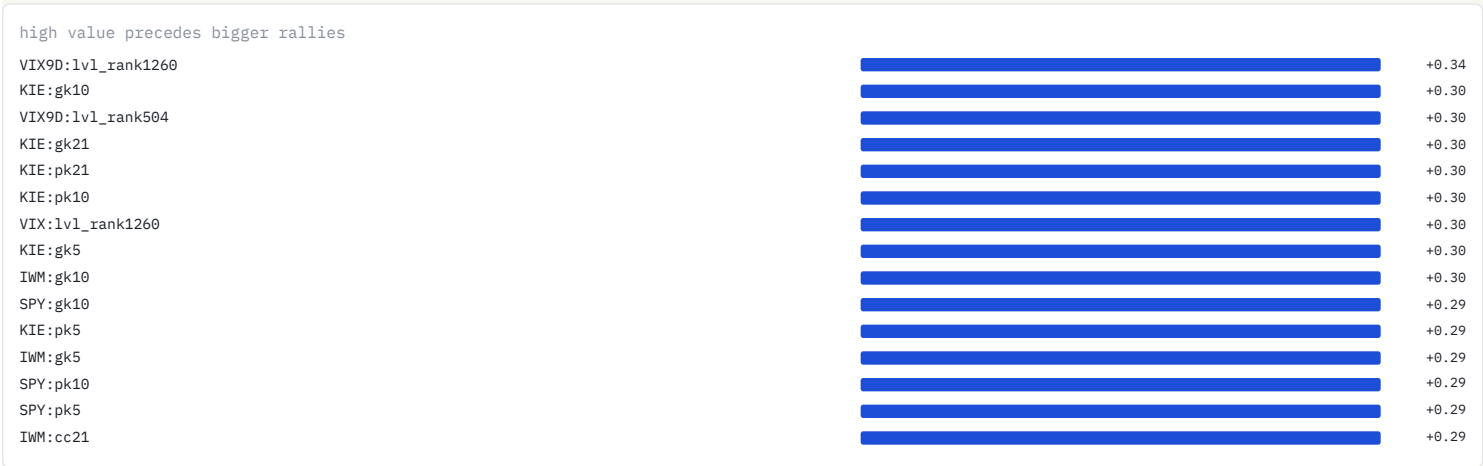
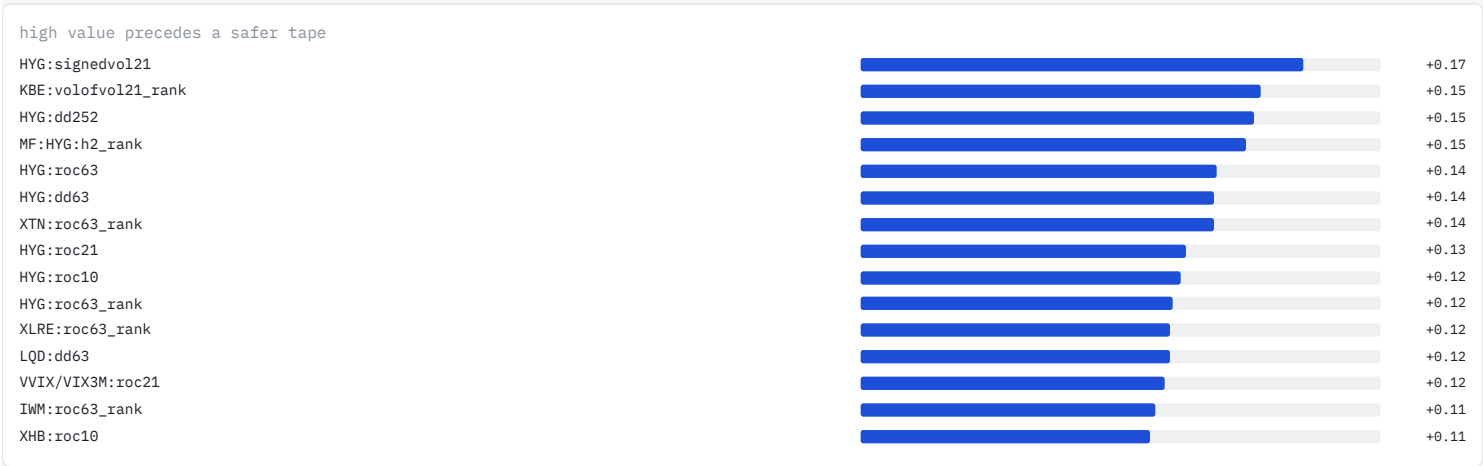
real pooled AUC vs the mean of 20 null pipelines (sd 0.056, max 0.622) · z 5.0

The permutation test shifts the label series by a random offset of at least a year and reruns the entire walk-forward pipeline, feature selection included; the resulting AUC is what selection and fitting produce from nothing.

WHY VERSION 1 WAS RETIRED: EVERY V1 RULE BY SAMPLE HALF

GROUP	V1 RULE	SIDE	FIRES	P ALL	P 2008-17	P 2018-26	FWD21 2018-26	HALVES
main	SETUP_rates	off	289	0.25	0.26	0.2	0.86	ok / FAIL
main	SETUP_complac	off	38	0.24	0.42	0.5	-1.74	ok / n<6
main	ONSET_impulse	off	62	0.27	0.35	0.23	0.82	ok / FAIL
main	ONSET_vvixlag	off	182	0.27	0.29	0.25	1.23	ok / ok
main	ONSET_movelag	off	14	0.43	0.5	0.33	0.18	ok / n<6
main	ONSET_ovxdiv	off	40	0.3	0.24	0.37	-0.19	FAIL / ok
main	CONT_2ndleg	off	64	0.39	0.41	0.39	1.69	ok / ok
main	CAP_alldims	on	52	0.15	0.19	0.1	3.5	ok / ok
main	CAP_vix_gvz	on	117	0.17	0.17	0.17	1.95	ok / ok
main	CAP_vvixout	on	9	0	0	0	4.4	ok / n<6
main	ONCONF_collapse	on	225	0.19	0.18	0.2	1.02	ok / FAIL
main	FAIL_yieldsup	off	56	0.39	0.44	0.36	-0.5	ok / ok
pair	HYG_credit_crack	off	7	0.71	0.67	1	-2.7	ok / n<6
pair	KRE_banks_vs_yields (removed)	off	35	0.46	0.61	0.27	1.45	ok / FAIL
pair	TLT_duration_bid_calm	off	6	0.5	0.75	0	0.83	n<6 / n<6
pair	XLU_defensive_bid	off	37	0.38	0.44	0.29	0.61	ok / ok
pair	XLRE_bid_yields_up	off	13	0.31		0.31	0.12	n<6 / FAIL
pair	IWM_beta_chase	off	53	0.3	0.36	0.16	1.79	ok / FAIL
pair	XLU_month_lead	off	197	0.31	0.33	0.36	0.29	ok / ok
pair	VVIX_confirms_spike	on	58	0.16	0.2	0.06	3.83	FAIL / ok
realized	RVX_premium_collapse	off	114	0.19	0.29	0.31	1.76	ok / ok
realized	SPY_realized_outruns_implied	off	48	0.29	0.5	0.31	2.49	ok / FAIL
realized	SPY_implied_outruns_realized	on	211	0.2	0.27	0.15	2.1	ok / ok
realized	QQQ_realized_outruns_VXN	off	41	0.27	0.39	0.21	1.95	ok / FAIL
realized	HYG_credit_vol_cheap_VIX	off	110	0.26	0.2	0.33	0.83	FAIL / ok
realized	XLE_energy_vol_cheap_OVX	off	100	0.23	0.19	0.29	-0.56	FAIL / ok
realized	TLT_MOVE_leads_realized	off	45	0.24	0.27	0.37	-0.95	FAIL / ok
realized	HYG_realized_expanding_spike	off	61	0.25	0.33	0.14	2.94	ok / FAIL
context	rates_pressure_vix_asleep	off	216	0.07	0.05	0.03	2.38	FAIL / FAIL
context	complacency_both_compressed	off	256	0.25	0.33	0.23	-0.02	ok / ok
context	vol_collapsing_from_high	on	250	0.25	0.43	0.28	0.8	FAIL / FAIL
context	vix_floor_vvix_floor	off	332	0.1	0.11	0.08	1.25	FAIL / FAIL
context	drawdown_no_capitulation	off	1297	0.37	0.43	0.3	2.44	ok / ok

FEATURES THAT CARRIED THE SAME SIGN IN EVERY ERA (TO 2012, 2013-19, 2020-26)



VOL EPISODES AND MULTIFRACTAL STATE

START	END	KIND	SESSIONS	SPY DD INSIDE %	SPY START TO END %
2026-06-23	2026-07-16	realized	17	-0.6	+2.3
2026-06-05	2026-06-15	implied	7	-1.7	+2.3
2026-03-31	2026-04-30	realized	22	0.0	+10.0
2026-02-05	2026-04-30	implied	59	-6.7	+6.2
2026-01-20	2026-01-22	implied	3	0.0	+1.7
2025-11-06	2025-12-03	implied	19	-2.7	+2.0

2025-10-10	2025-10-27	implied	12	0.0	+4.8
2025-08-01	2025-08-08	implied	6	0.0	+2.5
2025-03-05	2025-06-11	realized	69	-15.8	+3.4
2025-02-24	2025-07-10	implied	95	-18.2	+5.3
2025-01-10	2025-01-21	implied	7	0.0	+3.8
2025-01-07	2025-02-11	realized	24	-1.4	+2.8
2024-12-18	2024-12-24	implied	5	-0.0	+2.9
2024-10-07	2024-11-07	implied	24	0.0	+4.8

SERIES	H2	RANK	WIDTH	RANK	ALPHA R	RANK	VR5	VR21
SPY	0.420	0.185	0.299	0.754	0.774	0.377	0.902	0.863
HYG	0.389	0.058	0.363	0.919	0.830	0.752	0.799	0.776
VIX	0.328	0.097	0.230	0.065	0.620	0.004	0.632	0.358
QQQ	0.454	0.278	0.284	0.808	0.773	0.526	0.931	0.892
TMH	0.467	0.429	0.179	0.593	0.622	0.097	0.769	0.862

Episodes: realized = SPY 21-day realized vol crossing above its trailing-504 80th percentile, ending below the 50th; implied = VIX level rank likewise. Multifractal: rolling 504-session MF-DFA; h2 is the Hurst exponent of returns, width = h(-4) - h(4), alpha_abs the DFA exponent of absolute returns; vr5 and vr21 are Lo-MacKinlay variance ratios.

TAPE TODAY, AS TRAILING RANKS (0 = LOWEST OF THE LAST TWO YEARS, 1 = HIGHEST)

VIX LEVEL 0.14 trailing rank	VIX 5D ROC 0.28 trailing rank	VIX 21D ROC 0.42 trailing rank	VVIX/VIX 5D ROC 0.53 trailing rank
VIX/VIX3M 5D ROC 0.24 trailing rank	MOVE LEVEL 0.85 trailing rank	MOVE 5D ROC 0.18 trailing rank	10Y YIELD 21D CHANGE 0.96 trailing rank
SPY REALIZED 21D (%) 10.20 level	SPY PARKINSON 21D (%) 7.35 level	SPY REALIZED LEVEL 0.25 trailing rank	SPY RV 21/63 0.42 trailing rank
SPY PARKINSON / CLOSE-TO-CLOSE 0.24 trailing rank	VIX OVER REALIZED 0.53 trailing rank	VIX PREMIUM 5D CHANGE 0.34 trailing rank	SPY VOLUME VS 63D 0.06 trailing rank
SPY VOLATILITY OF VOLUME 0.36 trailing rank	SPY RANGE COMPRESSION 0.57 trailing rank	SPY ILLIQUIDITY 0.45 trailing rank	HYG PARKINSON VOL 0.83 trailing rank
HYG SIGNED VOLUME 21D -0.35 level	VIX OVER HYG REALIZED 0.19 trailing rank	XLV RELATIVE 63D 0.61 trailing rank	DISPERSION 5D 0.39 trailing rank
AVG CORRELATION 21D 0.30 trailing rank	AVG CORRELATION 5D CHANGE 0.65 trailing rank	REALIZED-VOL BREADTH 0.07 trailing rank	VOLUME BREADTH 0.50 trailing rank
COMPRESSION BREADTH 0.10 trailing rank	ROC BREADTH 21D 0.24 trailing rank	RSP/SPY 21D 0.06 trailing rank	RSP/SPY 63D 0.12 trailing rank

TICKER LAYER: BEHAVIOUR AROUND 76 VOL-EPIISODE STARTS AND UNDER HIGH DISPERSION

TICKER	REL ROC5	RANK	-5	-1	0	+3	+10	FWD 10D REL %	FWD 21D REL %	N
XES			0.43	0.31	0.32	0.3	0.48	-1.81	-1.95	55
XLC			0.38	0.32	0.29	0.51	0.47	-1.10	-0.88	26
BNO			0.41	0.61	0.63	0.53	0.48	-1.03	-2.57	50
KCE			0.44	0.35	0.28	0.28	0.36	-1.02	-0.51	60
XOP			0.44	0.39	0.36	0.31	0.6	-0.97	-2.12	55
XTN			0.51	0.46	0.41	0.43	0.38	-0.93	-0.33	50
IEF			0.49	0.77	0.91	0.88	0.38	-0.81	-1.83	64
KBE			0.47	0.49	0.43	0.41	0.38	-0.76	-1.27	60
XME			0.38	0.32	0.29	0.35	0.39	-0.74	-0.02	55
HYG			0.54	0.73	0.91	0.87	0.43	-0.71	-0.97	55
GLD			0.61	0.67	0.85	0.78	0.42	-0.57	+0.02	62
KIE			0.6	0.61	0.68	0.59	0.47	-0.41	-0.30	60
KRE			0.43	0.56	0.57	0.45	0.51	-0.37	-0.49	55
XLE			0.48	0.55	0.56	0.42	0.55	-0.35	-0.35	75
XHE			0.49	0.53	0.64	0.54	0.59	-0.28	+0.05	50
XLP			0.63	0.7	0.81	0.79	0.53	-0.26	+0.51	75
TLT			0.47	0.68	0.9	0.87	0.38	-0.21	-1.45	64
XLV			0.54	0.62	0.67	0.64	0.53	-0.20	+0.56	75
XLB			0.56	0.41	0.5	0.35	0.42	-0.20	+0.26	75
XBI			0.53	0.4	0.46	0.46	0.61	-0.16	+0.09	57
XRT			0.55	0.38	0.51	0.41	0.44	-0.12	-0.05	55
INM			0.48	0.42	0.39	0.51	0.54	-0.11	-0.18	69
RWR			0.55	0.63	0.71	0.65	0.53	-0.11	+0.60	65
XSW			0.42	0.3	0.31	0.29	0.54	-0.10	-0.12	50
XLRE			0.49	0.69	0.83	0.7	0.5	-0.08	-0.28	34
XLI			0.51	0.45	0.51	0.44	0.54	-0.05	+0.27	75
XLF			0.62	0.55	0.5	0.49	0.42	-0.03	+0.05	75
XLK			0.49	0.39	0.31	0.39	0.47	+0.09	-0.31	75
XLY			0.45	0.4	0.45	0.47	0.5	+0.18	-0.10	75
QQQ			0.5	0.37	0.29	0.26	0.5	+0.21	+0.03	74
XHS			0.52	0.58	0.66	0.61	0.62	+0.28	+0.96	50
XLU			0.62	0.77	0.8	0.72	0.54	+0.29	+0.93	75
XTL			0.54	0.4	0.35	0.53	0.59	+0.32	+0.92	50
XHB			0.44	0.44	0.41	0.51	0.46	+0.38	-0.27	57
XPH			0.59	0.58	0.56	0.57	0.61	+0.39	+0.93	55
XSD			0.49	0.33	0.23	0.36	0.56	+0.57	+0.71	57
XAR			0.48	0.4	0.42	0.38	0.6	+0.74	+0.74	50

TICKER	LOW DISPERSION	MID	HIGH DISPERSION
QQQ	+0.08	+0.14	+0.08
TLT	-0.63	-0.64	-0.47
IEF	-0.67	-0.68	-0.60
HYG	-0.43	-0.41	-0.45
INM	+0.13	+0.07	0.00
GLD	-0.29	-0.34	-0.39
XLC	+0.11	0.00	-0.14
XLY	+0.13	+0.03	-0.05
XLP	-0.25	-0.15	-0.02
XLE	-0.06	-0.10	+0.03
XLF	+0.03	-0.05	-0.15
XLV	-0.07	-0.03	-0.09
XLI	+0.07	+0.05	+0.08
XLB	-0.14	+0.07	+0.10
XLRE	-0.47	-0.27	-0.24
RWR	+0.01	-0.03	-0.02
XAR	+0.12	+0.18	+0.32
KBE	-0.28	-0.28	-0.45
XBI	-0.28	-0.01	+0.42
KCE	-0.10	-0.01	-0.29
XHE	+0.03	-0.03	0.00
XHS	+0.04	0.00	+0.06
XHB	-0.21	-0.29	-0.16
KIE	-0.06	-0.05	-0.21
XME	-0.12	-0.12	+0.01
XES	+0.08	-0.21	-0.37
XOP	+0.05	-0.30	+0.15
XPH	-0.12	+0.04	-0.07
KRE	-0.31	-0.33	-0.35
XRT	-0.05	-0.05	-0.14
XSD	+0.40	+0.26	+0.38
XSW	+0.13	+0.12	+0.34
XTL	-0.26	-0.11	-0.03
XTN	-0.10	-0.19	+0.03
XLK	+0.04	+0.11	+0.17
XLU	-0.09	-0.01	+0.27
BNO	-0.01	-0.11	-0.51

Method: features are rates of change, ratios and trailing-504 percentile ranks built from daily OHLCV (close-to-close, Parkinson and Garman-Klass realized vol; volume and the volatility of volume; range compression; cross-sectional dispersion and correlation; implied over realized; equal-weight breadth; MF-DFA spectra). Models: 15 features chosen inside each training window by same-

signed rank correlation in both halves, depth-2 gradient-boosted trees constrained to be monotone in that sign, refit each January on data whose labels end before the test year. Thresholds are trailing quantiles of past out-of-sample values. KRE-based features are excluded from the candidate set by the user's direction. Nothing here assumes a distribution.